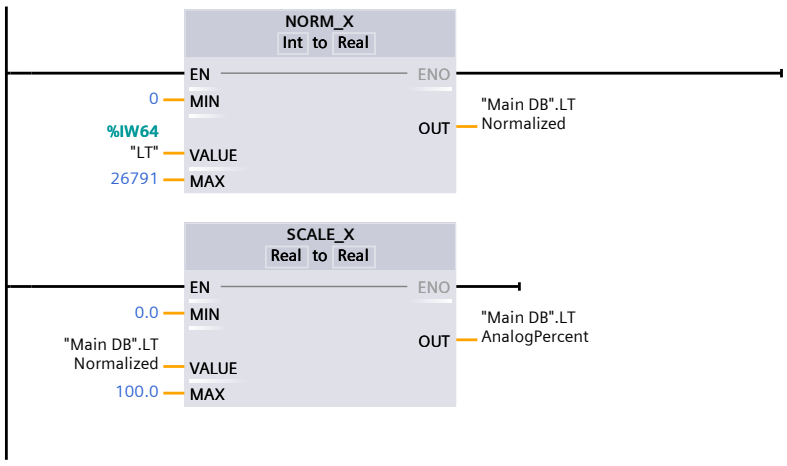
Program blocks

Main [OB1]

Main Properties							
General							
Name	Main	**Number**	1	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

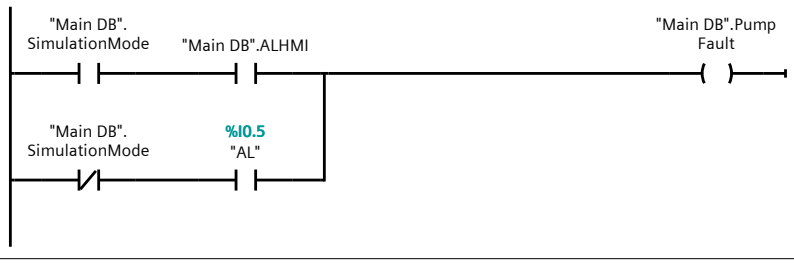
Network 1: LT Analog Input Scaling



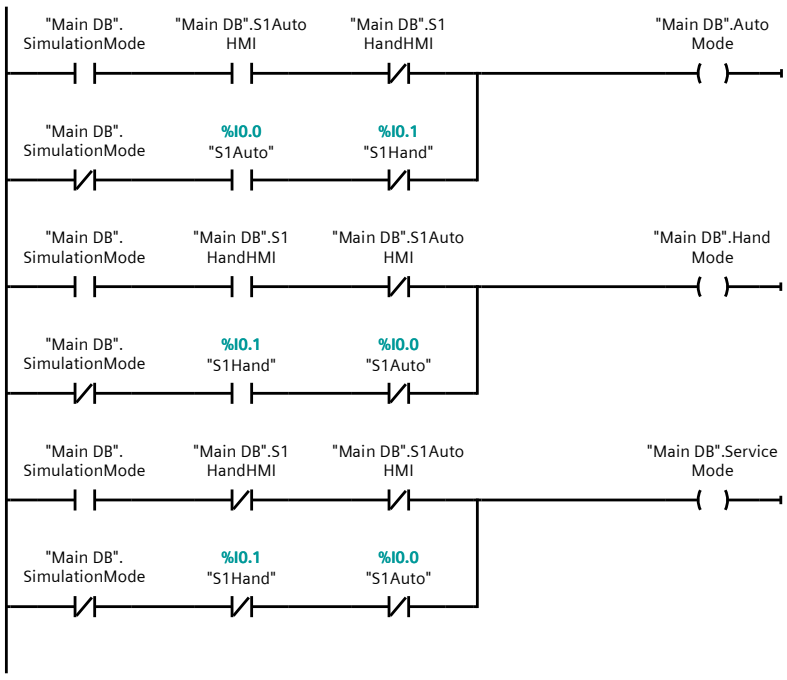
Network 2: LT Source Selection and Level Conversion

```
0001 IF "Main DB".SimulationMode THEN
0002     "Main DB".LevelPercent := "Well DB"."TankLevel%";
0003     "Main DB".LevelInLiters := "Well DB".TankLevel;
0004 ELSE
0005     "Main DB".LevelPercent := "Main DB".LTAnalogPercent;
0006     "Main DB".LevelInLiters := ("Main DB".LevelPercent / 100.0) * 1250.0;
0007 END_IF;
0008
0009 "Main DB".LTMeters := ("Main DB".LevelPercent * 2.5) / 100.0;
0010
```

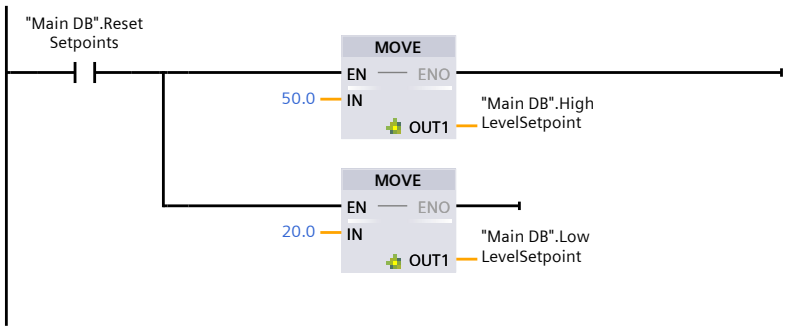
Network 3: Pump Fault Detection



Network 4: Operating Mode Selection



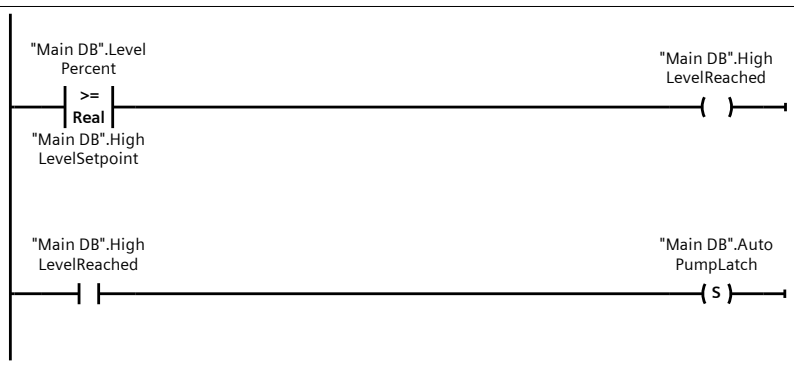
Network 5: HMI Default Settings



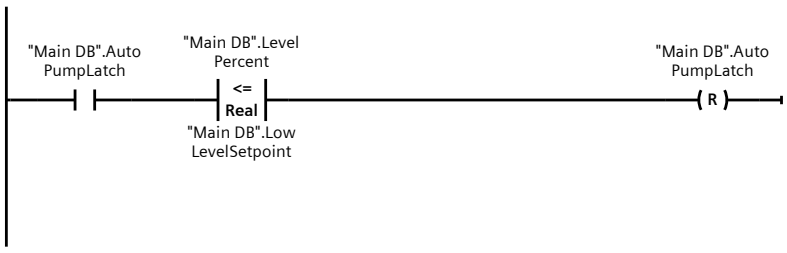
Network 6: HMI Setpoint Validation

```
0001
0002 IF "Main DB".LowLevelSetpoint <> "Main DB".LowSetpointPrevious THEN
0003
0004     "Main DB".LowLevelSetpoint := LIMIT(
0005                                       MN := 0.0,
0006                                       IN := "Main DB".LowLevelSetpoint,
0007                                       MX := "Main DB".HighLevelSetpoint - 5.0
0008     );
0009
0010 ELSIF "Main DB".HighLevelSetpoint <> "Main DB".HighSetpointPrevious THEN
0011
0012     "Main DB".HighLevelSetpoint := LIMIT(
0013                                       MN := "Main DB".LowLevelSetpoint + 5.0,
0014                                       IN := "Main DB".HighLevelSetpoint,
0015                                       MX := 100.0
0016     );
0017
0018 END_IF;
0019
0020 "Main DB".HighSetpointPrevious := "Main DB".HighLevelSetpoint;
0021 "Main DB".LowSetpointPrevious := "Main DB".LowLevelSetpoint;
0022
```

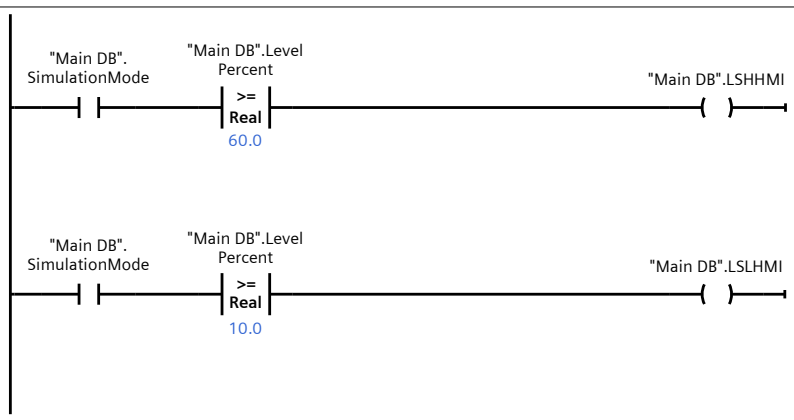
Network 7: High Level Setpoint Detection



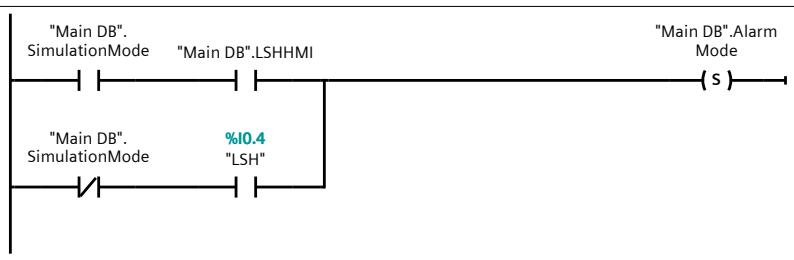
Network 8: Low Level Setpoint Reset



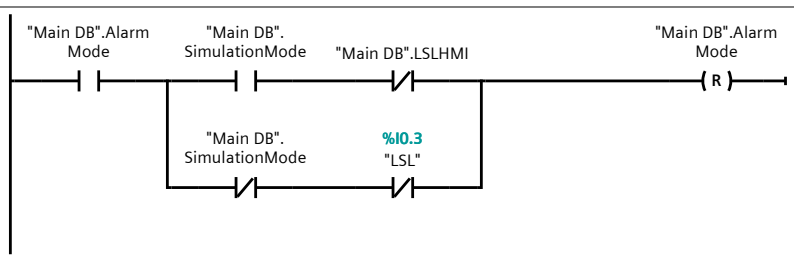
Network 9: Float Switch Simulation



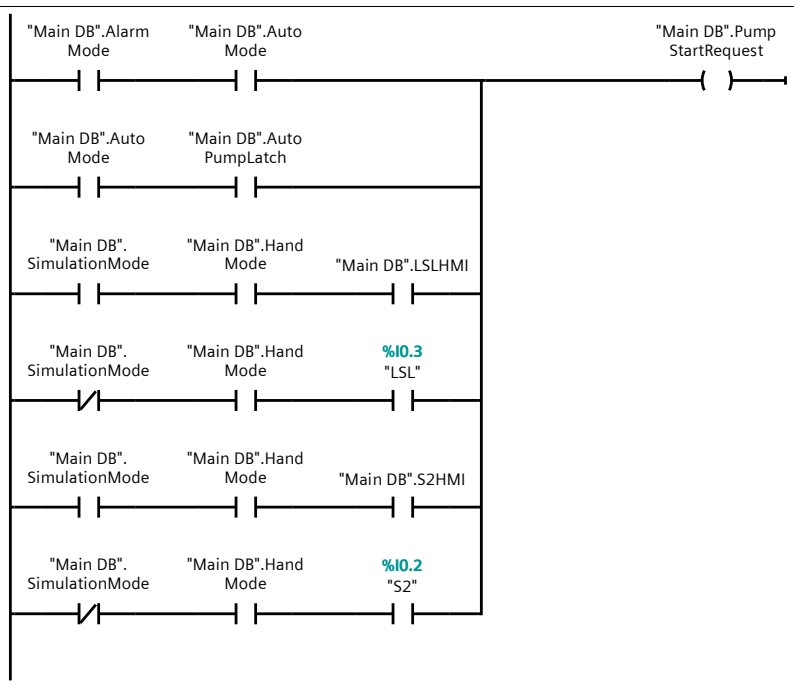
Network 10: High Level Alarm Activation



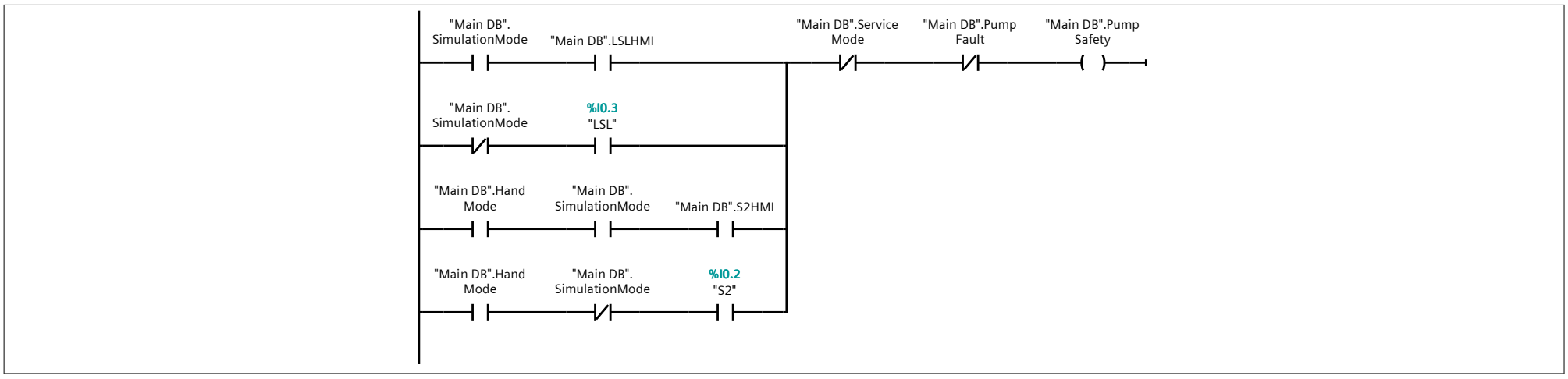
Network 11: Alarm Reset at Low Level



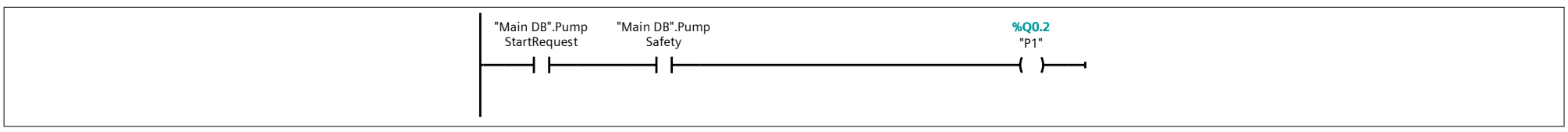
Network 12: Pump Start Request



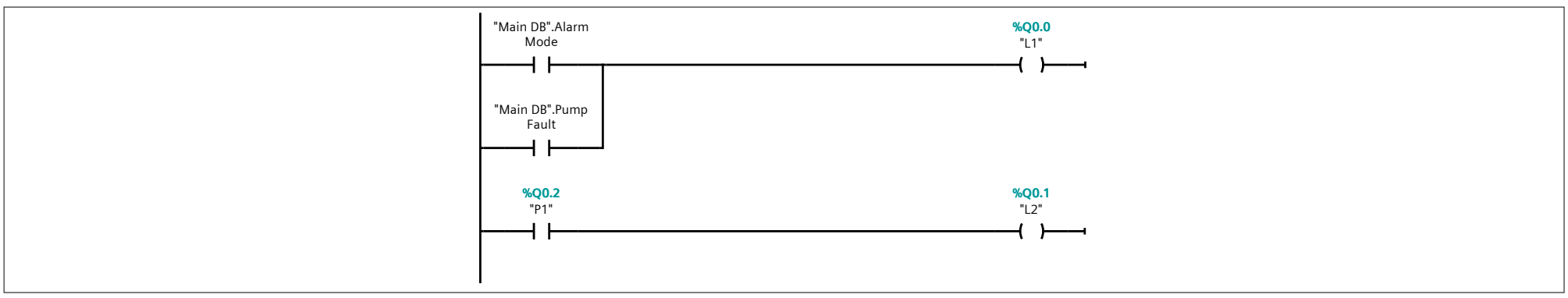
Network 13: Pump Safety Interlocks



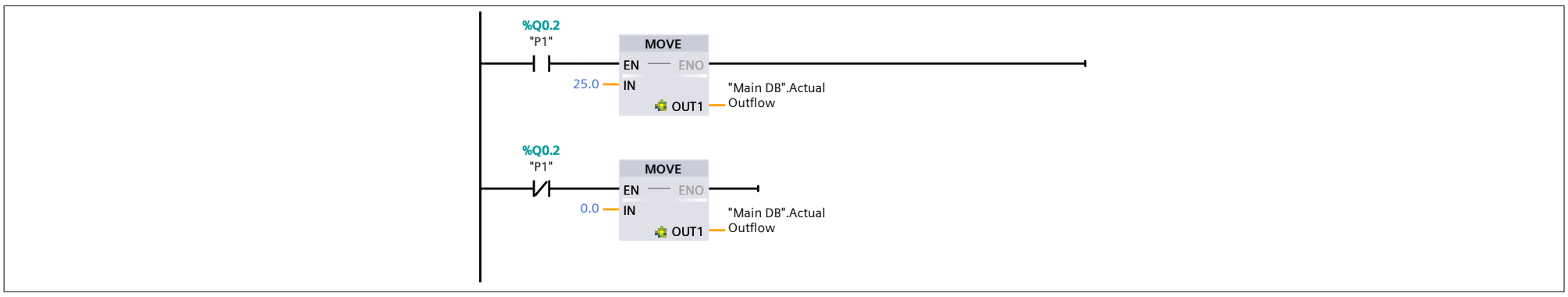
Network 14: Pump Output Control



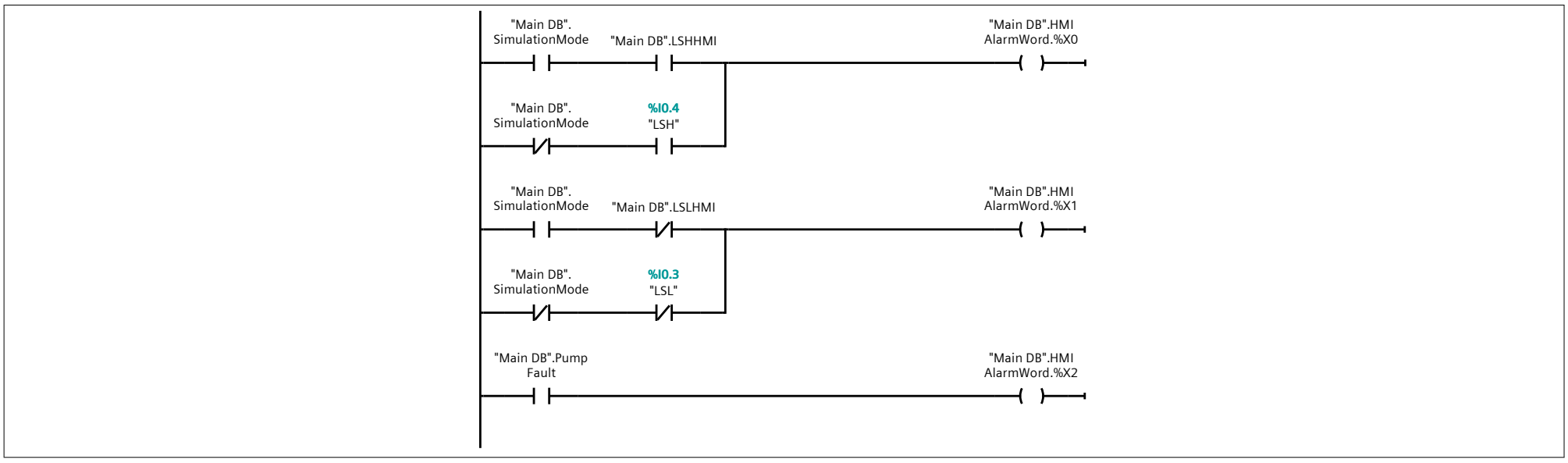
Network 15: Pump Status Indicators



Network 16: Pump Outflow Calculation



Network 17: HMI Alarm Word



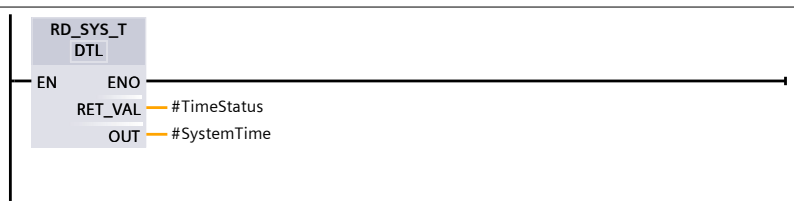
Program blocks

Random Inflow Generator [FB1]

Random Inflow Generator Properties							
General							
Name	Random Inflow Generator	**Number**	1	**Type**	FB	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
▼ Output									
RandomInflow	Real	0.0	Non-retain	True	True	True	False		
InOut									
▼ Static									
TimeStatus	Int	0	Non-retain	True	True	True	False		
▼ SystemTime	DTL	DTL#1970-01-01-00:00:00	Non-retain	True	True	True	False		
YEAR	UInt	1970	Non-retain	True	True	True	False		
MONTH	USInt	1	Non-retain	True	True	True	False		
DAY	USInt	1	Non-retain	True	True	True	False		
WEEKDAY	USInt	5	Non-retain	True	True	True	False		
HOUR	USInt	0	Non-retain	True	True	True	False		
MINUTE	USInt	0	Non-retain	True	True	True	False		
SECOND	USInt	0	Non-retain	True	True	True	False		
NANOSECOND	UDInt	0	Non-retain	True	True	True	False		
RandomValue	Int	0	Non-retain	True	True	True	False		
Temp									
Constant									

Network 1: System Time Acquisition



Network 2: Random Value Generation

```
0001 IF #TimeStatus = 0 THEN
0002     #RandomValue :=
0003     UDINT_TO_INT(
0004                   #SystemTime.NANOSECOND MOD UDINT#3
0005     ) + 1;
0006 END_IF;
```

Network 3: Random Inflow Rate Selection

```
0001 IF #RandomValue = 1 THEN
0002     #RandomInflow := 10.0;
0003 ELSIF #RandomValue = 2 THEN
0004     #RandomInflow := 15.0;
0005 ELSE
0006     #RandomInflow := 7.0;
0007 END_IF;
0008
```

Program blocks

Random Generator DB [DB1]

Random Generator DB Properties							
General							
Name	Random Generator DB	**Number**	1	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
▼ Output									
RandomInflow	Real	0.0	False	True	True	True	False		
InOut									
▼ Static									
TimeStatus	Int	0	False	True	True	True	False		
▼ SystemTime	DTL	DTL#1970-01-01-00:00:00	False	True	True	True	False		
YEAR	UInt	1970	False	True	True	True	False		
MONTH	USInt	1	False	True	True	True	False		
DAY	USInt	1	False	True	True	True	False		
WEEKDAY	USInt	5	False	True	True	True	False		
HOUR	USInt	0	False	True	True	True	False		
MINUTE	USInt	0	False	True	True	True	False		
SECOND	USInt	0	False	True	True	True	False		
NANOSECOND	UDInt	0	False	True	True	True	False		
RandomValue	Int	0	False	True	True	True	False		

Program blocks

Main DB [DB4]

Main DB Properties							
General							
Name	Main DB	**Number**	4	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Static									
RandomInflow	Real	0.0	False	True	True	True	False		
ActualInflow	Real	0.0	False	True	True	True	False		
ActualOutflow	Real	0.0	False	True	True	True	False		
S1AutoHMI	Bool	false	False	True	True	True	False		
S1HandHMI	Bool	false	False	True	True	True	False		
S2HMI	Bool	false	False	True	True	True	False		
AutoMode	Bool	false	False	True	True	True	False		
HandMode	Bool	false	False	True	True	True	False		
ServiceMode	Bool	false	False	True	True	True	False		
LevelPercent	Real	0.0	False	True	True	True	False		
HighLevelReached	Bool	false	False	True	True	True	False		
AutoPumpLatch	Bool	false	False	True	True	True	False		
LSHHMI	Bool	false	False	True	True	True	False		
LSLHMI	Bool	false	False	True	True	True	False		
AlarmMode	Bool	false	False	True	True	True	False		
HighLevelSetpoint	Real	50.0	False	True	True	True	False		
LowLevelSetpoint	Real	20.0	False	True	True	True	False		
ResetSetpoints	Bool	false	False	True	True	True	False		
AlarmHelp	Bool	false	False	True	True	True	False		
LTMeters	Real	0.0	False	True	True	True	False		
ALHMI	Bool	false	False	True	True	True	False		
PumpFault	Bool	false	False	True	True	True	False		
HMIAlarmWord	Word	16#0	False	True	True	True	False		
HighSetpointPrevious	Real	50.0	False	True	True	True	False		
LowSetpointPrevious	Real	20.0	False	True	True	True	False		
LTNormalized	Real	0.0	False	True	True	True	False		
LTAnalogPercent	Real	0.0	False	True	True	True	False		
SimulationMode	Bool	false	False	True	True	True	False		
PumpStartRequest	Bool	false	False	True	True	True	False		
PumpSafety	Bool	false	False	True	True	True	False		
LevelInLiters	Real	0.0	False	True	True	True	False		

Program blocks

Well Simulation [FB2]

Well Simulation Properties							
General							
Name	Well Simulation	**Number**	2	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
▼ Input									
CycleTime	Real	0.1	Non-retain	True	True	True	False		
Inflow	Real	0.0	Non-retain	True	True	True	False		
Outflow	Real	0.0	Non-retain	True	True	True	False		
Output									
InOut									
▼ Static									
TankLevel	Real	0.0	Non-retain	True	True	True	False		
ActualInflow	Real	0.0	Non-retain	True	True	True	False		
ActualOutflow	Real	0.0	Non-retain	True	True	True	False		
TankLevel%	Real	0.0	Non-retain	True	True	True	False		
Temp									
Constant									

```
0001 #ActualOutflow := #Outflow * #CycleTime;
0002 #ActualInflow := #Inflow * #CycleTime;
0003
0004 #TankLevel := #TankLevel + (#ActualInflow - #ActualOutflow);
0005
0006 IF #TankLevel > 1250.0 THEN
0007   #TankLevel := 1250.0;
0008 END_IF;
0009
0010 IF #TankLevel < 0.0 THEN
0011   #TankLevel := 0.0;
0012 END_IF;
0013
0014 #"TankLevel%" := (#TankLevel / 1250.0) * 100.0;
0015
```

Symbol	Address	Type	Comment
#"TankLevel%"		Real	
#ActualInflow		Real	
#ActualOutflow		Real	
#CycleTime		Real	
#Inflow		Real	
#Outflow		Real	
#TankLevel		Real	

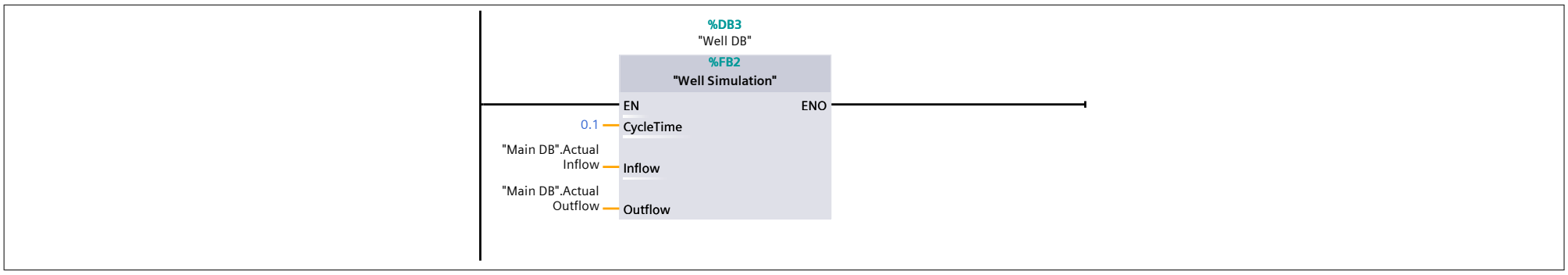
Program blocks

Well Cycle [OB30]

Well Cycle Properties							
General							
Name	Well Cycle	**Number**	30	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Event_Count	Int		Events discarded
Temp			
Constant			

Network 1: Well Level Simulation



Program blocks

Well DB [DB3]

Well DB Properties							
General							
Name	Well DB	**Number**	3	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
▼ Input									
CycleTime	Real	0.1	False	True	True	True	False		
Inflow	Real	0.0	False	True	True	True	False		
Outflow	Real	0.0	False	True	True	True	False		
Output									
InOut									
▼ Static									
TankLevel	Real	0.0	False	True	True	True	False		
ActualInflow	Real	0.0	False	True	True	True	False		
ActualOutflow	Real	0.0	False	True	True	True	False		
TankLevel%	Real	0.0	False	True	True	True	False		

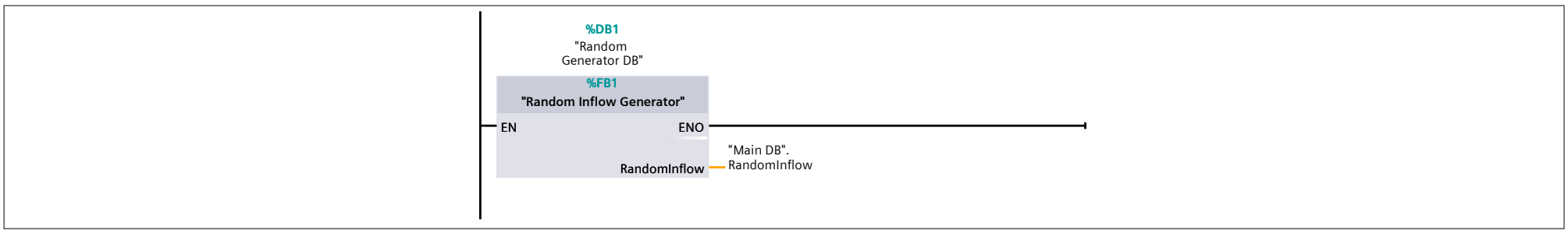
Program blocks

Inflow Generator [OB123]

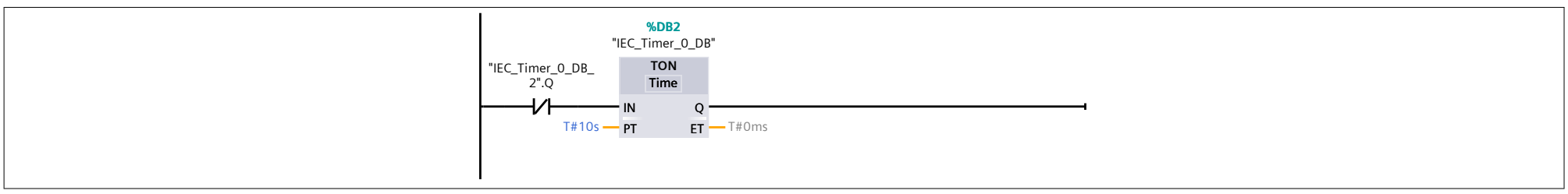
Inflow Generator Properties							
General							
Name	Inflow Generator	**Number**	123	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

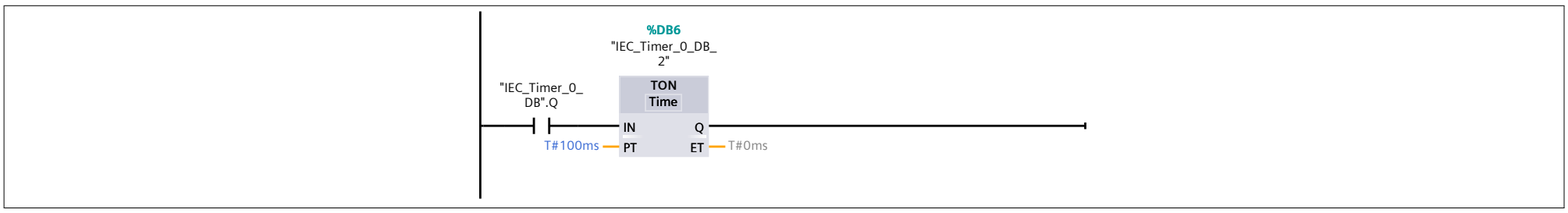
Network 1: Random Inflow Generator Call



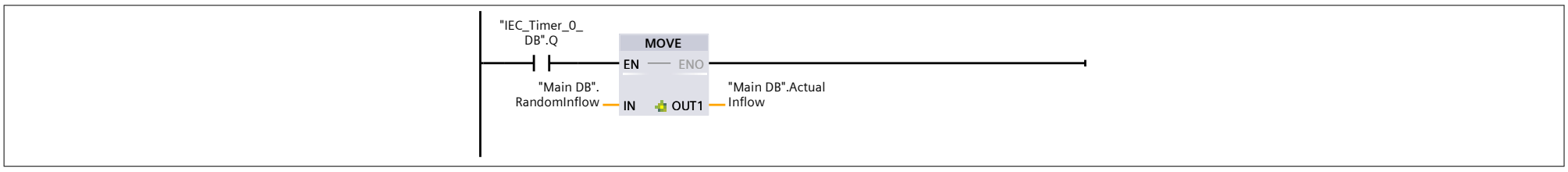
Network 2: Inflow Update Interval Timer



Network 3: Inflow Update Pulse Generation



Network 4: Actual Inflow Update



Program blocks / System blocks / Program resources

IEC_Timer_0_DB [DB2]

IEC_Timer_0_DB Properties							
General							
Name	IEC_Timer_0_DB	**Number**	2	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	Simatic	**Comment**		**Family**	IEC
Version	1.0	**User-defined ID**	IEC_TMR				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Static									
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		

Program blocks / System blocks / Program resources

IEC_Timer_0_DB_2 [DB6]

IEC_Timer_0_DB_2 Properties							
General							
Name	IEC_Timer_0_DB_2	**Number**	6	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	Simatic	**Comment**		**Family**	IEC
Version	1.0	**User-defined ID**	IEC_TMR				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Static									
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		